

The weak $(p, 1)$ limit case fails

Lane 19 packet for arXiv:1409.6476

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Source and target

The source is Kati Ain and Eve Oja, *On (p, r) -null sequences and their relatives*, arXiv:1409.6476 [1]. In Proposition 4.5 they prove, for $1 \leq p < \infty$ and $1 < r \leq p^*$ with $r < \infty$ if $p = 1$, that

$$\mathcal{W}_{(p,r)} = \mathcal{W}_{(p,r)} \circ \mathcal{W} \quad \text{and} \quad \mathbf{w}_{(p,r)} = \mathcal{W}_{(p,r)}(\mathbf{w}).$$

Remark 4.6 asks whether this remains true in the limit case $r = 1$.

of $\mathcal{W}_p$ in [27, pp. 20–21] and of $\mathcal{K}_{(p,r)}$ (see Section 2.1), we get a natural factorization $T = \overline{\Phi}_{(x_n)} S$ with $(x_n) \in \ell_p^w(X)$, where $\overline{\Phi}_{(x_n)}$ is the injective associate of $\Phi_{(x_n)}$ and $S \in \mathcal{L}(Y, Z)$, where $Z := \ell_r / \ker \Phi_{(x_n)}$. Since $\Phi_{(x_n)} \in \mathcal{W}_{(p,r)}(\ell_r, X)$, we have $\overline{\Phi}_{(x_n)} \in \mathcal{W}_{(p,r)}^{\text{sur}}(Z, X) = \mathcal{W}_{(p,r)}(Z, X)$, because $\mathcal{W}_{(p,r)}$ is surjective. Since ℓ_r is reflexive, also Z is, and therefore $S \in \mathcal{W}(Y, Z)$. This proves that $\mathcal{W}_{(p,r)} = \mathcal{W}_{(p,r)} \circ \mathcal{W}$. Now, using this, we have

$$\mathbf{w}_{(p,r)} = \mathcal{W}_{(p,r)}(\mathbf{b}) = (\mathcal{W}_{(p,r)} \circ \mathcal{W})(\mathbf{b}) = \mathcal{W}_{(p,r)}(\mathcal{W}(\mathbf{b})) = \mathcal{W}_{(p,r)}(\mathbf{w}). \quad \square$$

Remark 4.6. We do not know whether Proposition 4.5 holds in the “limit” case $r = 1$, i.e., for $\mathcal{W}_{(p,1)}$. It does not hold in the other “limit” case $p = 1$, $r = \infty$, i.e., for $\mathcal{W}_1 = \mathcal{W}_{(1,\infty)}$. Indeed, as we saw above, $\mathcal{W}_1(c_0, c_0) = \mathcal{L}(c_0, c_0)$, and hence

$$\mathbf{w}_1(c_0) = \mathcal{W}_1(\mathbf{b})(c_0) = \mathbf{b}(c_0) \neq \mathbf{w}(c_0) = \mathcal{W}_1(\mathbf{w}).$$

In particular, $\mathcal{W}_{(1,\infty)} \not\subset \mathcal{W}$. In all other cases $\mathcal{W}_{(p,r)} \subset \mathcal{W}$. For $r \neq 1$, this is clear from Proposition 4.5. But $\mathcal{W}_{(p,1)} \subset \mathcal{W}_{(p,r)}$ (by the definition of $\mathcal{W}_{(p,\cdot)}$, because $B_{\ell_1} \subset B_{\ell_r}$).

Remark 4.7. In the case $p = 1$, $1 \leq r \leq p^*$, including also the case $p = 1$, $r = \infty$ (cf. Remark 4.6), Proposition 4.5 holds in a strong form for a large class of Banach spaces X . Namely, for X that does not contain c_0 isomorphically. In this case (and only in this case), $\ell_1^w(X) = \ell_1^u(X)$, by the

Theorem 4.8 gives an omnibus characterization of weakly (p, r) -null sequences under the same non-limit hypotheses. Remark 4.9 asks whether Theorem 4.8 holds in the limit cases, including $r = 1$.

(p, r) -conv(w_k). We also have an $M > 0$ such that $\|y_n\| \leq M$ for all $n \in \mathbb{N}$. Now $(z_k) := (Mw_k) \in \ell_p(X)$ and $x_n \in (p, r)$ -conv(z_k) for all $n \in \mathbb{N}$. As (x_n) is weakly null in X , for every $x^* \in X^*$ and $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $|x^*(x_n)| \leq \varepsilon$ for all $n \geq N$. Hence, (x_n) is uniformly weakly (p, r) -null.

The implication (e) $\Rightarrow$ (a) is clear from the definitions. $\square$

Remark 4.9. As we saw, all implications of Theorem 4.8, except (b) $\Rightarrow$ (c), also hold in the “limit” cases $r = 1$ and $p = 1$, $r = \infty$. In the proof, we used that the implication (b) $\Rightarrow$ (c) is immediate from Proposition 4.5 (see also Remark 4.6). We do not know whether Theorem 4.8 holds in these cases. If $p = 1$ and $1 \leq r \leq p^*$, Theorem 4.8 holds in a stronger form for those Banach spaces X that do not contain c_0 isomorphically. Indeed, by Remark 4.7, in condition (b), “weakly $(1, r)$ -compact” is the same as “unconditionally $(1, r)$ -compact” and in condition (c) “weakly $\mathcal{W}_{(1,r)}$ -compact” is the same as “ $\mathcal{U}_{(1,r)}$ -compact” and also the same as “ $\mathcal{N}_{(\infty, \infty, r^*)}$ -compact”. In condition (d), “weakly $\mathcal{W}_{(1,r)}$ -null” is the same as “weakly $\mathcal{U}_{(1,r)} \circ \mathcal{K}$ -null”, which is the same as “ $\mathcal{U}_{(1,r)}$ -null”, since compact operators take weakly null sequences to null sequences, i.e., $\mathcal{K} \subset \mathcal{V}$ (see, e.g., [22, 1.11.4]). This shows that in the special case when $p = 1$, $1 \leq r \leq p^*$, and X does not contain c_0 isomorphically, all conditions of Theorem 4.1 are equivalent to the conditions of Theorem 4.8.

This packet gives a negative answer for $r = 1$, for every $1 \leq p < \infty$.

Definitions used

For a sequence $(z_k) \in \ell_p^w(X)$, the weak $(p, 1)$ -convex hull is

$$(p, 1)\text{-conv}(z_k) = \left\{ \sum_{k=1}^{\infty} a_k z_k : (a_k) \in B_{\ell_1} \right\}.$$

An operator $T : Y \rightarrow X$ is weakly $(p, 1)$ -compact if $T(B_Y)$ is contained in such a hull for some $(z_k) \in \ell_p^w(X)$.

Proof intuition

The endpoint $r = 1$ is special because the natural coefficient space is ℓ_1 , whose quotients have the Schur property. Thus every weakly $(p, 1)$ -compact operator is completely continuous: after the standard factorization through a quotient of ℓ_1 , weakly null sequences become norm-null. Therefore an operator in $\mathcal{W}_{(p,1)} \circ \mathcal{W}$ maps bounded sets first to weakly compact sets and then, by complete continuity, to norm-compact sets. So every operator in the product ideal is compact. But the canonical inclusion $\ell_1 \hookrightarrow c_0$ is weakly $(p, 1)$ -compact and not compact.

Lemma 1. *For every $1 \leq p < \infty$, every weakly $(p, 1)$ -compact operator is completely continuous.*

Proof. Let $T : Y \rightarrow X$ be weakly $(p, 1)$ -compact. Then there is $(z_k) \in \ell_p^w(X)$ such that

$$T(B_Y) \subseteq \Phi(B_{\ell_1}), \quad \Phi(a) = \sum_{k=1}^{\infty} a_k z_k.$$

The map $\Phi : \ell_1 \rightarrow X$ is bounded because (z_k) is weakly p -summable and hence bounded. Let $q : \ell_1 \rightarrow Z = \ell_1 / \ker \Phi$ be the quotient map and let $\bar{\Phi} : Z \rightarrow X$ be the induced injective operator. Since $T(B_Y) \subseteq \Phi(B_{\ell_1})$, T takes values in $\text{ran } \Phi$, and the formula

$$Sy = q(a) \quad \text{whenever } \Phi a = Ty$$

defines a bounded linear map $S : Y \rightarrow Z$ with $T = \bar{\Phi}S$. Indeed, for $\|y\| \leq 1$ one may choose $a \in B_{\ell_1}$ with $\Phi a = Ty$, so $\|Sy\| \leq 1$ in the quotient norm.

The space Z has the Schur property, because it is a quotient of ℓ_1 . If $y_n \rightarrow 0$ weakly in Y , then $Sy_n \rightarrow 0$ weakly in Z , hence $\|Sy_n\| \rightarrow 0$. Therefore

$$\|Ty_n\| = \|\bar{\Phi}Sy_n\| \rightarrow 0.$$

So T is completely continuous. □

Lemma 2. *For every $1 \leq p < \infty$,*

$$\mathcal{W}_{(p,1)} \circ \mathcal{W} \subseteq \mathcal{K}.$$

Proof. Let $T = AB$ where $B : Y \rightarrow Z$ is weakly compact and $A : Z \rightarrow X$ is weakly $(p, 1)$ -compact. By Lemma 1, A is completely continuous.

The set $B(B_Y)$ is relatively weakly compact. By Eberlein–Smulian, every sequence in $B(B_Y)$ has a weakly convergent subsequence. If $z_{n_j} \rightarrow z$ weakly in Z , then $z_{n_j} - z \rightarrow 0$ weakly, hence $Az_{n_j} \rightarrow Az$ in norm by complete continuity. Thus $A(B(B_Y)) = T(B_Y)$ is relatively norm sequentially compact, hence relatively norm compact. Therefore T is compact. □

Theorem 1. *For every $1 \leq p < \infty$,*

$$\mathcal{W}_{(p,1)} \neq \mathcal{W}_{(p,1)} \circ \mathcal{W}.$$

Proof. Let $j : \ell_1 \rightarrow c_0$ be the canonical inclusion. We first show that $j \in \mathcal{W}_{(p,1)}$.

Let (e_k) be the unit vector basis of c_0 . Since $(c_0)^* = \ell_1$ and $\ell_1 \subseteq \ell_p$ for $1 \leq p < \infty$, the sequence (e_k) belongs to $\ell_p^w(c_0)$. Moreover,

$$j(B_{\ell_1}) = \left\{ \sum_{k=1}^{\infty} a_k e_k : (a_k) \in B_{\ell_1} \right\} = (p, 1)\text{-conv}(e_k).$$

Thus j is weakly $(p, 1)$ -compact.

But j is not compact, since $(je_k) = (e_k)$ has no norm-convergent subsequence in c_0 . Lemma 2 says that every operator in $\mathcal{W}_{(p,1)} \circ \mathcal{W}$ is compact. Hence $j \notin \mathcal{W}_{(p,1)} \circ \mathcal{W}$, proving the strict inequality. □

Corollary 1. *Theorem 4.8 of the source does not extend to the limit case $r = 1$. More precisely, for every $1 \leq p < \infty$, condition (b) of Theorem 4.8 does not imply condition (c) when $r = 1$.*

Proof. In $X = c_0$, take $x_n = e_n$. The sequence (e_n) is weakly null in c_0 , because every functional on c_0 is an ℓ_1 sequence. As above, $(e_k) \in \ell_p^w(c_0)$ and $e_n \in (p, 1)\text{-conv}(e_k)$ for every n . Thus (e_n) is weakly null and relatively weakly $(p, 1)$ -compact, which is condition (b) of Theorem 4.8 with $r = 1$.

Suppose condition (c) held. Then $\{e_n : n \in \mathbb{N}\}$ would be weakly $\mathcal{W}_{(p,1)}$ -compact: there would be a weakly $(p, 1)$ -compact operator $A : Y \rightarrow c_0$ and a relatively weakly compact set $K \subseteq Y$ with

$$\{e_n : n \in \mathbb{N}\} \subseteq A(K).$$

By Lemma 1, A is completely continuous, so the same Eberlein–Smulian argument as in Lemma 2 shows that $A(K)$ is norm compact. This is impossible because the set $\{e_n : n \in \mathbb{N}\}$ has no norm-convergent subsequence in c_0 . □

Verification and limitations

The counterexample is in ZFC and works for every $1 \leq p < \infty$. It answers the source's $r = 1$ limit question negatively. It does not address any positive endpoint behavior under extra hypotheses, such as the source's $\text{no-}c_0$ hypothesis in Remark 4.9.

Bounded novelty checks were performed on 2026-06-20 against the local run indexes, the local parsed arXiv corpus, and web searches for exact and close phrases around Proposition 4.5, Theorem 4.8, $r = 1$, and weakly $(p, 1)$ -compact operators. These checks found the source paper and no prior answer to the endpoint question.

References

- [1] K. Ain and E. Oja, *On (p, r) -null sequences and their relatives*, arXiv:1409.6476, 2014.
- [2] R. E. Megginson, *An Introduction to Banach Space Theory*, Graduate Texts in Mathematics 183, Springer, 1998.